科學 HW 7

1. The B.V.P $-u''(x)=f(x)$ with $u(0)=u(1)=0$, $f(x)=\frac{1}{x}$

By the formula in (12.3), $u(x)=\int_0^1 G(x,s)f(s)\,ds$, where $G(x,s)=\begin{cases} s(1-x) & \text{if } 0\le s\le x \\ x(1-s) & \text{if } x\le s\le 1\end{cases}$

$\therefore u(x)=\int_0^1 G(x,s)\cdot\frac{1}{s}\,ds=\int_0^x \frac{s(1-x)}{s}ds+\int_x^1\frac{x(1-s)}{s}ds=(1-x)\cdot x+x\int_x^1 \frac{1}{s}-1\,ds$

$=(1-x)x+x\left[\log s-s\Big|_x^1\right]=(1-x)x+x\left[-1-\log x+x\right]=(-\log x)x$ #

2. From the identity $w_{j+1}v_{j+1}-w_jv_j=(w_{j+1}-w_j)v_j+(v_{j+1}-v_j)w_{j+1}$. $\forall w_h, v_h\in V_h$, we obtain

$\sum_{j=0}^{h-1}(w_{j+1}-w_j)v_j=w_hv_h-w_0v_0-\sum_{j=0}^{h-1}(v_{j+1}-v_j)w_{j+1}$ ...... (*)

L_h: the operator $(L_hw_h)(x_j)=-\frac{(w_{j+1}-2w_j+w_{j-1})}{h^2}$.

let $w_{-1}=v_{-1}=0$, $\forall w_h, v_h\in V_h^0$, we obtain:

$(L_hw_h,v_h)_h=-h\sum_{j=0}^{h-1}\left(\frac{(w_{j+1}-w_j)-(w_j-w_{j-1})}{h^2}\right)v_j=-h^{-1}\sum_{j=0}^{h-1}\left[(w_{j+1}-w_j)-(w_j-w_{j-1})\right]v_j$

$=-h^{-1}\left\{\sum_{j=0}^{h-1}[(w_{j+1}-w_j)]v_j-\sum_{j=0}^{h-1}[(w_j-w_{j-1})]v_j\right\}$

$=-h^{-1}\left\{\sum_{j=0}^{h-1}[(w_{j+1}-w_j)]v_j-\left[\sum_{j=0}^{h-2}[(w_{j+1}-w_j)]v_{j+1}+(w_0-w_{-1})v_0\right]\right\}$

$=-h^{-1}\sum_{j=0}^{h-1}(w_{j+1}-w_j)v_j+h^{-1}\left[(w_0-w_{-1})v_0+\sum_{j=0}^{h-2}(w_{j+1}-w_j)v_{j+1}\right]$

$\overset{\text{by }(*)}{=}-h^{-1}\sum_{j=0}^{h-1}(w_{j+1}-w_j)(v_{j+1}-v_j)$

Taking $v_h=w_h$, we have $(L_hv_h,v_h)_h=-h^{-1}\sum_{j=0}^{h-1}(v_{j+1}-v_j)^2$ #

3. We know the $G^k\in V_h^0$ as the solution $L_hG^k=e^k$. $e^k(x_j)=\delta_{kj}$

For $j<k$, $G(x_j,x_k)=x_j(1-x_k)$ linear on the interval containing x_{j-1},x_j,x_{j+1} $\Rightarrow$ ($[0,x_k]$ and $[x_k,1]$)

$\therefore G_{j+1}-2G_j+G_{j-1}=0$, $(L_hG)_j=0$ for all $j\ne k$ (It's similar for $j>k$)

Consider $G_k=G(x_k,s)=s(1-s)$; $G_{k+1}=G(x_{k+1},s)=s(1-s)-sh$; $G_{k-1}=G(x_{k-1},s)=s(1-s)-h(1-s)$.

$\therefore G_{k+1}-2G_k+G_{k-1}=[s(1-s)-sh]-2s(1-s)+[s(1-s)-h(1-s)]=-h$

$\therefore (L_hG)_k=-\frac{G_{k+1}-2G_k+G_{k-1}}{h^2}=-\frac{(-h)}{h^2}=\frac{1}{h}$

So we have shown $L_hG=\frac{1}{h}e^k\Rightarrow L_h(hG)=e^k$, $\therefore G^k(x_j)=hG(x_j,x_k)$ #

(or $G^k(x_k)=hG(x_k,x_j)$)